

Olympiad Test Paper — Science (Class 7)

Chapter 8: Winds, Storms and Cyclones

Subject	Science (NCERT Class 7)
Chapter	8 — Winds, Storms and Cyclones
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Remember the two big ideas of this chapter: air flows from a region of higher pressure to a region of lower pressure, and faster-moving air exerts lower pressure. No calculator is needed. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

1. Which statement about air is correct?

- (A) Air has no weight and exerts no pressure.
- (B) Air exerts pressure only downward.
- (C) Air exerts pressure only at sea level.
- (D) Air has weight and exerts pressure in all directions.

2. Wind is best described as:

- (A) air that is being heated by the Sun
- (B) water vapour rising into the sky
- (C) air moving from a region of higher pressure to a region of lower pressure
- (D) air moving from a region of lower pressure to a region of higher pressure

3. A little water is boiled in a tin can, the cap is screwed on tightly, and the can is then cooled. The can gets crushed because:

- (A) the cold water outside pushes the walls in
- (B) the metal shrinks and tears
- (C) the air and steam inside cool, their pressure drops, and the higher outside air pressure crushes it
- (D) cooling creates extra air inside that pushes the walls

4. At a beach on a hot afternoon you feel a cool wind blowing from the sea toward the land. This sea breeze happens because:

- (A) the sea heats up faster than the land
- (B) the land heats up faster than the sea, so warm air over the land rises and cooler sea air moves in

- (C) the land is always cooler than the sea
- (D) sea water evaporates and pushes the air toward land

5. You hold two inflated balloons hanging side by side a few centimetres apart and blow a stream of air through the gap between them. The balloons will:

- (A) move apart
- (B) stay exactly where they are
- (C) both rise upward
- (D) move toward each other

6. Inside the towering clouds of a thunderstorm, the chief reason for the violent activity is:

- (A) cold dry air sinking quickly to the ground
- (B) lightning being created first and then pulling up the air
- (C) rapidly rising warm, moist air together with falling water drops
- (D) the ground heating the clouds from below by touch

7. Air over the equator and air over the poles are heated very differently by the Sun. The direct result of this uneven heating is that:

- (A) large-scale wind currents are set up that move heat around the planet
- (B) there is no wind anywhere on Earth
- (C) the poles become hotter than the equator
- (D) air stops having any weight near the equator

8. As a cyclone forms over a warm ocean, water vapour in the rising air condenses into droplets. This condensation helps the cyclone grow because it:

- (A) cools the air and stops the wind
- (B) makes the air heavier so it sinks back down
- (C) releases heat, which warms the air further, lowers the pressure more, and pulls in even faster wind
- (D) adds weight that raises the pressure at the centre

9. Wind blows fast over the curved top of a roof while the air inside the house is still.

During a storm a poorly tied roof tends to be lifted off because:

- (A) the rain makes the roof float
- (B) the fast air above the roof has lower pressure than the still air below, so the roof is pushed up
- (C) the still air below has lower pressure and sucks the roof down
- (D) wind always pushes roofs straight down

10. The calm, clear, low-pressure region at the very centre of a cyclone is called the:

- (A) eye
- (B) eye wall
- (C) storm surge
- (D) front

11. Two identical sealed tins are taken to the top of a tall mountain and to the seashore.

Compared with the seashore tin, the mountain-top tin experiences:

- (A) greater outside air pressure
- (B) lower outside air pressure, because there is less air above it
- (C) exactly the same outside air pressure
- (D) no air pressure at all

12. On a hot day the air just above a sun-baked open field tends to:

- (A) sink, because hot air is heavier (B) stay perfectly still
- (C) rise, because it expands, becomes lighter, and is pushed up
- (D) turn into water

13. The single greatest cause of death and destruction along the coast when a cyclone makes landfall is usually the:

- (A) storm surge — a huge rise of sea water flooding the low coast
- (B) lightning strikes (C) loud thunder
- (D) hailstones

14. A balloon is stretched over the mouth of an empty bottle, and the bottle is then placed in a bowl of hot water. The balloon inflates because:

- (A) hot water leaks into the bottle (B) the glass squeezes the air upward
- (C) the air inside warms, expands, and pushes the balloon outward
- (D) the balloon makes its own air

15. At night the wind at a coast usually blows from the land toward the sea (a land breeze) because:

- (A) the land stays warmer than the sea all night (B) the sea cools faster than the land
- (C) the land cools faster than the sea, so now the warmer air is over the sea and rises, drawing air out from the land
- (D) winds always reverse at sunset for no reason

16. When a cyclone alert is issued for a coastal area, the safest immediate action is to:

- (A) go to the beach to watch the waves
- (B) move to a safer place such as a cyclone shelter and follow official instructions
- (C) ignore it unless the wind is already strong
- (D) drive toward the sea to escape inland traffic

17. The flash of lightning and the rumble of thunder are produced at the same instant, yet we usually see the lightning before we hear the thunder. This is because:

- (A) thunder happens a few seconds after lightning
- (B) light travels much faster than sound (C) sound travels faster than light
- (D) the eye reacts before the ear by several seconds

18. India's summer monsoon brings heavy rain mainly because, in summer, the moisture-laden winds blow:

- (A) from the sea toward the land (B) from the land toward the sea
- (C) straight downward from the sky (D) from the poles toward the equator

19. A strip of paper is held just below your lower lip and you blow hard straight across its top surface. The free end of the strip will:

- (A) be pushed straight down
- (B) not move at all
- (C) lift up
- (D) curl toward your mouth

20. A cyclone gathers strength over warm tropical seas but quickly weakens after it moves onto land. The main reason it weakens over land is that:

- (A) the land is colder than space
- (B) buildings physically block the wind
- (C) the eye closes up on contact with soil
- (D) it is cut off from the warm, moist sea air that fed it

21. The narrow ring of the most violent, fastest winds and heaviest rain in a cyclone is the:

- (A) eye wall
- (B) eye
- (C) calm centre
- (D) outer edge, where the storm is mildest

22. Air pressure pushes on us from every side, yet we are not crushed. The best explanation is that:

- (A) the pressure of the fluids and air inside our body pushes outward and balances it
- (B) air pressure is too weak to be felt at all
- (C) our skin is completely airtight and stiff
- (D) gravity cancels the air pressure exactly

23. The instrument used to measure the speed of the wind is the:

- (A) thermometer
- (B) barometer
- (C) anemometer
- (D) rain gauge

24. Why does air rush in to take the place of warm air that has risen above a hot region?

- (A) because the cooler air is attracted by sunlight
- (B) because cooler air is lighter than warm air
- (C) because the Sun pulls the air sideways
- (D) because the rising warm air leaves a region of lower pressure behind it

25. During a severe cyclone, people are advised **not** to drink water from open, flooded sources mainly because:

- (A) the flood water is likely contaminated and can spread disease
- (B) flood water is too cold
- (C) the water has too much salt to taste good
- (D) open water reflects lightning

26. By day the air pressure over the warm land is lower than over the cooler sea. Knowing wind blows from high to low pressure, the daytime breeze must blow:

- (A) from the land to the sea
- (B) from the sea to the land
- (C) straight up only
- (D) in a full circle with no net direction

27. A sealed plastic bottle is filled with hot air and capped, then put in a refrigerator for an hour. When taken out it is found dented inward. The dent appears because:

- (A) the fridge squeezed it mechanically
- (B) cold makes plastic expand and fold
- (C) the cooled air inside lost pressure, so the outside air pressed the sides in
- (D) water from the fridge soaked into the plastic

28. A coastal town keeps getting accurate cyclone warnings a day or two in advance. The technology most responsible for this early warning is:

- (A) the sundial
- (B) the household thermometer
- (C) weather satellites and radar
- (D) the kitchen pressure cooker

29. Lightning is essentially:

- (A) sunlight reflected off rain
- (B) burning gas in the upper air
- (C) sound turning into light
- (D) a huge electric spark caused by charges built up in storm clouds

30. Coastal regions of India have a more moderate, less extreme daily temperature than far-inland regions largely because:

- (A) water heats and cools more slowly than land, so sea breezes keep the coast milder
- (B) the sea has no effect on coastal climate
- (C) inland air has no weight
- (D) the coast receives no sunlight

31. A train standing on a platform is about to be passed by a fast express. A person standing very close to the platform edge can be dangerously pulled toward the moving train because:

- (A) the fast-moving train pushes air against the person
- (B) the fast air between the person and the train has lower pressure, so the higher pressure behind the person pushes them in
- (C) the train magnetically attracts people
- (D) the rails vibrate and pull people forward

32. Which statement correctly compares a tornado with a cyclone?

- (A) a tornado is always far larger than a cyclone
- (B) a tornado has no spinning motion
- (C) tornadoes and cyclones are exactly the same size
- (D) a tornado is generally a much smaller, narrower, very violent whirling column of air

33. In the self-feeding cycle of a growing cyclone, which sequence is correct?

- (A) pressure rises → wind slows → cyclone weakens
- (B) cold air sinks → it warms the sea → wind stops
- (C) moist air rises → vapour condenses → heat released → pressure falls → faster wind rushes in → more moist air rises
- (D) lightning forms → it heats the sea → the eye widens

34. Recorded wind speeds in the eye wall of a strong cyclone can reach roughly:

- (A) 5 to 10 km/h
- (B) 30 to 40 km/h
- (C) the same as a gentle breeze
- (D) 150 to 250 km/h

35. Two neighbouring regions of the atmosphere have a large pressure difference between them. Compared with two regions whose pressures are nearly equal, the wind between the first pair will be:

- (A) weaker
- (B) stronger
- (C) exactly the same
- (D) absent

36. An anemometer often has small cups mounted on arms that spin in the wind. The **faster** the cups spin, the:

- (A) lower the wind speed
- (B) higher the air temperature
- (C) higher the wind speed
- (D) heavier the rainfall

37. The sea breeze and the land breeze both occur because:

- (A) the Moon pulls the air toward the coast
- (B) land and water heat up and cool down at different rates
- (C) sea water is saltier than land
- (D) the coast is always windier than inland

38. A weather report says the air pressure over a region has suddenly **fallen** sharply. This most likely signals:

- (A) an approaching storm or cyclone, because storms are low-pressure systems
- (B) clear, calm, settled weather ahead
- (C) that the wind will completely stop
- (D) that the temperature has dropped to freezing

39. Why does a cyclone's storm surge cause such severe flooding of low-lying coasts?

- (A) the surge is only rainwater collecting slowly
- (B) the sea level drops, exposing dry land
- (C) the surge is made entirely of fresh river water
- (D) the very low pressure and fierce winds heap up a great mound of sea water that washes far inland

40. A glass tumbler is filled to the brim with water, a stiff card is held over the mouth, and the tumbler is quickly turned upside down. The card and water stay in place because:

- (A) the air pressure pushing up on the card is greater than the weight of the water pushing down
- (B) the water is glued to the card
- (C) water cannot fall when upside down
- (D) the glass holds the water by magnetism

41. On a calm, very hot and humid summer afternoon, conditions are ripe for a thunderstorm because such weather provides:

- (A) cold, dry, sinking air
- (B) strong steady wind from the poles
- (C) low humidity and clear skies for days
- (D) plenty of warm, moisture-rich air that can rise quickly

42. During India's **winter** months the prevailing winds over much of the country tend to blow:

- (A) from the sea toward the land, bringing heavy rain
- (B) straight upward

- (C) from the equator toward the poles, bringing snow
- (D) from the land toward the sea, so they are mostly dry

43. Which of these is a sound safety measure to take when a cyclone warning is in force?

- (A) keep all doors and windows wide open to let the wind pass through
- (B) store drinking water and keep emergency supplies and a radio ready
- (C) go outside to take photographs of the storm
- (D) shelter under a tall lone tree

44. Wind blows hard across the open mouth of a chimney at the top of a house. As a result, smoke inside is drawn **up** the chimney more strongly. This is because the fast wind across the top:

- (A) raises the pressure at the chimney mouth
- (B) blows fresh air down into the room
- (C) has no effect on the smoke
- (D) lowers the pressure at the chimney mouth, so the higher pressure below pushes the smoke up

45. Put the early life of a cyclone in the right order: (i) wind spirals inward, gaining speed (ii) the Sun heats the ocean and warm moist air rises (iii) water vapour condenses, releasing heat and dropping the pressure further. The correct order is:

- (A) (i) → (ii) → (iii)
- (B) (iii) → (ii) → (i)
- (C) (ii) → (iii) → (i)
- (D) (ii) → (i) → (iii)

46. The very low pressure at the centre of a cyclone is important because it:

- (A) is what draws the surrounding air rushing inward as fierce wind
- (B) keeps the wind from blowing at all
- (C) makes the sea level rise far out at sea only
- (D) warms the upper atmosphere and dries it out

47. Which set of features correctly describes a cyclone from centre outward?

- (A) violent eye → calm eye wall → still outer ring
- (B) high-pressure eye → dry eye wall → no clouds
- (C) calm low-pressure eye → violent eye wall → outer spiralling clouds
- (D) calm eye → calm eye wall → calm edge

48. A region of the atmosphere is being warmed strongly from below. Over time this region tends to develop:

- (A) higher pressure, because warm air is heavier
- (B) lower pressure, because the warm air expands and rises away
- (C) no change in pressure
- (D) zero air pressure

49. A weather station reports that its anemometer cups are spinning very slowly today. This tells us that:

- (A) the wind is light today
- (B) the air is very hot
- (C) heavy rain is falling
- (D) the air pressure is extremely high

- 50.** Why does the sea breeze typically arrive only **during the day** and not at night?
- (A) the sea is only warm at night
 - (B) wind cannot blow after sunset
 - (C) only during the day does the land heat faster than the sea, creating the needed pressure difference
 - (D) the Sun blows the wind directly toward shore
- 51.** Before a cyclone strikes, fishermen are warned not to go out to sea chiefly because:
- (A) the violent winds and high waves can capsize their boats
 - (B) the fish swim away during storms
 - (C) sea water becomes too cold to sail in
 - (D) cyclones make the sea water disappear
- 52.** A small ball is held in the upward stream of air from a hair-dryer pointed straight up; the ball hovers and even resists being pushed sideways out of the stream. The ball stays in the stream because:
- (A) the air is sticky and glues the ball
 - (B) the ball is weightless inside the stream
 - (C) the dryer's heat lifts the ball by magnetism
 - (D) the fast-moving air in the stream has lower pressure, so the slower air around pushes the ball back in
- 53.** The modern reason coastal communities suffer fewer cyclone deaths than they did a century ago is mostly that:
- (A) cyclones have become much weaker over time
 - (B) the sea no longer floods the coast
 - (C) people have stopped living near the coast entirely
 - (D) accurate satellite warnings now give people time to evacuate to shelters
- 54.** Which correctly orders the events of a thunderstorm developing into rain?
- (A) rain falls first, then warm air rises, then clouds form
 - (B) warm moist air rises and cools → water vapour forms tall clouds → drops grow heavy and fall as rain
 - (C) lightning strikes the ground → clouds appear → air warms
 - (D) cold air sinks → the sea heats → rain rises into the sky
- 55.** The whole pattern of winds and the monsoon over India is ultimately driven by:
- (A) the Moon's gravity alone
 - (B) the uneven heating of the Earth's land, sea and atmosphere by the Sun
 - (C) ocean currents with no role for the Sun
 - (D) the spinning of household fans
- 56.** In the bottle-and-balloon activity, after the hot-water bath the same bottle is moved into a bowl of ice water. The balloon will now:
- (A) get sucked inward / collapse into the bottle as the air inside cools and its pressure drops
 - (B) inflate even more
 - (C) stay exactly as it was in the hot water

(D) pop off and float away

57. As a cyclone intensifies over the sea, the pressure at its centre and the speed of its winds change in which way?

- (A) the centre pressure rises and the winds get faster
- (B) the centre pressure falls lower and the winds get faster
- (C) the centre pressure falls and the winds get slower
- (D) both the pressure and the wind speed stay constant

58. A coastal town has a steady sea breeze every afternoon. On one unusual afternoon the land is actually cooler than the sea. That afternoon you would expect:

- (A) an even stronger than usual sea breeze
- (B) the wind to blow straight upward
- (C) the usual sea breeze to weaken or reverse, since the needed pressure difference is gone or flipped
- (D) no change, because sea breezes never depend on temperature

59. Houses in cyclone-prone coastal areas are sometimes built with strong, well-anchored roofs rather than light loose ones mainly to resist the fact that fast wind over a roof:

- (A) raises the pressure above the roof and presses it down safely
- (B) cools the roof and cracks it
- (C) has no measurable effect on roofs
- (D) lowers the pressure above the roof, creating an upward lift that can tear it off

60. The eye of a cyclone is calm and clear, while the eye wall just around it is the most violent part. A ship that passes through a cyclone may therefore experience:

- (A) steady mild wind the entire time
- (B) fierce wind, then a sudden calm, then fierce wind again from the other side
- (C) no wind at all throughout
- (D) calm, then violent, then permanently calm

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 8: Winds, Storms and Cyclones

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	A	31	B	41	D	51	A
2	C	12	C	22	A	32	D	42	D	52	D
3	C	13	A	23	C	33	C	43	B	53	D
4	B	14	C	24	D	34	D	44	D	54	B
5	D	15	C	25	A	35	B	45	C	55	B
6	C	16	B	26	B	36	C	46	A	56	A
7	A	17	B	27	C	37	B	47	C	57	B
8	C	18	A	28	C	38	A	48	B	58	C
9	B	19	C	29	D	39	D	49	A	59	D
10	A	20	D	30	A	40	A	50	C	60	B

Answer-key distribution: A = 14, B = 15, C = 17, D = 14.

Worked Solutions

- (D)** Air is matter: it has mass (weight) and, like any fluid, it presses equally in every direction — up, down and sideways. It does not press only downward or only at sea level.
- (C)** Wind is simply moving air, and air always flows from where the pressure is higher to where it is lower. Heating by itself is not wind; the *movement* it sets up is.
- (C)** Boiling drives out most air and fills the can with steam; capping it traps low-density hot gas. On cooling, the steam condenses and the trapped gas contracts, so the inside pressure drops well below the outside air pressure, which then crushes the can. (The walls are pushed in by air, not by cold water or shrinking metal.)
- (B)** Land heats faster than water under the same Sun. The hot air over the land expands, becomes lighter and rises, lowering the pressure there; cooler, higher-pressure air from over the sea flows in to replace it — a sea breeze.
- (D)** The fast air stream you blow between the balloons has lower pressure than the still air on their outer sides. The higher outside pressure pushes the balloons together, so they move toward each other. (The common guess "move apart" is the trap.)

- 6. (C)** A thunderstorm's energy comes from warm, moisture-laden air shooting upward while water drops fall back down through it. Lightning is a *result* of this churning, not its cause; cold sinking air would suppress a storm.
- 7. (A)** The equator receives far more direct sunlight than the poles. This temperature difference creates pressure differences, which drive large-scale wind currents that carry heat from hot regions toward cold ones.
- 8. (C)** When water vapour condenses it releases its stored (latent) heat. That extra warmth makes the rising air even lighter, drops the central pressure further, and pulls in faster wind carrying still more moist air — a self-strengthening loop.
- 9. (B)** Fast wind streaming over the curved roof has lower pressure than the still air trapped inside the house. The greater pressure below then pushes the roof upward, so a loosely tied roof can be lifted off.
- 10. (A)** The calm, clear, low-pressure centre of a cyclone is the eye. The violent ring around it is the eye wall.
- 11. (B)** Air pressure comes from the weight of the air column above you. On a mountain there is less air overhead, so the outside pressure is lower than at the seashore.
- 12. (C)** Sun-heated air over a hot field expands, grows less dense (lighter) and is pushed up by the cooler, denser air around it. Hot air rises; it does not sink or sit still.
- 13. (A)** The deadliest part of a landfalling cyclone is usually the storm surge — a great heap of sea water driven onto the low coast, drowning huge areas. Lightning, thunder and hail cause far less of the destruction.
- 14. (C)** Warm water heats the air inside the bottle; the air expands and, having nowhere else to go, pushes out into the balloon and inflates it. No water enters and the balloon makes no air of its own.
- 15. (C)** At night the land loses heat faster than the sea, so now the sea is the warmer surface. Air over the sea rises, lowering pressure there, and cooler air flows out from the land toward the sea — a land breeze.
- 16. (B)** The right response to a cyclone alert is to move to a safe place such as a cyclone shelter and follow the authorities' instructions. Going to the beach or toward the sea, or ignoring the warning, is dangerous.
- 17. (B)** Light and sound are made together in a lightning stroke, but light travels enormously faster than sound, so the flash reaches our eyes well before the thunder reaches our ears.

- 18. (A)** In summer the land heats more than the sea, drawing moist air inland from the sea toward the land; that sea-to-land flow carries the rain of the summer monsoon.
- 19. (C)** Blowing across the top of the strip makes the air there move fast, so its pressure drops below the still air underneath. The higher pressure below then lifts the free end up.
- 20. (D)** A cyclone is fed by warm, moist air from a warm sea. Once it moves over land it loses that supply of warm moist air, so it weakens. (Buildings and "cold land" are not the main reason.)
- 21. (A)** The eye wall is the narrow ring just around the eye, where the winds are fastest and the rain heaviest — the most violent part of the cyclone.
- 22. (A)** The fluids and air inside our body push outward with about the same pressure as the atmosphere pushes inward, so the forces balance and we are not crushed. Air pressure is real and strong; gravity does not cancel it.
- 23. (C)** Wind speed is measured with an anemometer. (A thermometer measures temperature, a barometer pressure, a rain gauge rainfall.)
- 24. (D)** When warm air rises away from a hot region it leaves lower pressure behind. Surrounding higher-pressure air then rushes in to fill that low-pressure region. Cooler air is denser, not lighter, so options B and C are wrong.
- 25. (A)** Flood water during a cyclone is usually contaminated with sewage and debris and can spread disease, so people are told not to drink from open flooded sources.
- 26. (B)** Wind flows from high pressure to low pressure. Here the sea has the higher pressure and the land the lower, so the daytime breeze blows from the sea to the land.
- 27. (C)** Cooling the trapped air shrinks it and lowers its pressure below the outside air pressure, which then presses the bottle's sides inward, denting it. (The fridge does not squeeze it mechanically.)
- 28. (C)** Modern early warnings come from weather satellites and radar, which spot and track cyclones a day or two ahead. A sundial, thermometer or pressure cooker cannot do this.
- 29. (D)** Lightning is a giant electric spark: charges build up inside towering storm clouds and then jump as a huge discharge. It is not reflected sunlight or burning gas.
- 30. (A)** Water heats and cools much more slowly than land, so the nearby sea moderates coastal temperatures and the daily sea/land breezes keep the coast milder than the far interior.

- 31. (B)** The fast train drags the air between it and the person, making that air move quickly and so lowering its pressure. The higher pressure on the person's far side then pushes them toward the train — which is why one must stand well back.
- 32. (D)** A tornado is generally a much smaller, narrower but extremely violent whirling column of air, while a cyclone is a far larger system. Both spin, so "no spinning" and "same size" are wrong.
- 33. (C)** The growing-cyclone loop runs: moist air rises → vapour condenses → heat is released → central pressure falls → faster wind rushes in → it brings still more moist air, repeating the cycle. Rising pressure or sinking cold air would stop a cyclone, not build one.
- 34. (D)** In a strong cyclone the eye-wall winds can reach roughly 150 to 250 km/h — far above a gentle breeze.
- 35. (B)** The bigger the pressure difference between two regions, the harder the air is pushed from high to low — so a large pressure difference gives a stronger wind.
- 36. (C)** An anemometer's cups are spun by the wind: the faster they spin, the higher the wind speed. (It does not measure temperature or rainfall.)
- 37. (B)** Both the sea breeze and the land breeze arise because land and water heat up and cool down at different rates, creating the pressure differences that drive the breezes.
- 38. (A)** Storms and cyclones are low-pressure systems, so a sharp, sudden fall in air pressure usually warns that a storm is approaching — not that calm, settled weather is on the way.
- 39. (D)** A cyclone's very low central pressure and fierce winds pile up a great mound of sea water (the storm surge) that washes far inland, flooding low coasts. It is not slow rainwater or river water, and the sea does not retreat.
- 40. (A)** The atmosphere pushes up on the card with a pressure equal to the weight of many metres of air — far more than the weight of the small column of water in the glass — so the card and water are held in place.
- 41. (D)** Thunderstorms need plenty of warm, moisture-rich air that can rise quickly. A hot, humid afternoon supplies exactly that. Cold dry or low-humidity air would not build a storm.
- 42. (D)** In winter the land cools more than the sea, so winds tend to blow from the land out toward the sea. Because they come off the dry land, the winter winds are mostly dry — the opposite of the summer monsoon.
- 43. (B)** A sound cyclone precaution is to store drinking water and keep emergency supplies and a radio ready. Leaving windows wide open, going out to photograph the

storm, or sheltering under a lone tall tree are all unsafe.

44. (D) Fast wind across the chimney mouth lowers the pressure there. The higher pressure of the still air inside the house then pushes the smoke up and out through the chimney.

45. (C) A cyclone begins with the Sun heating the ocean so warm moist air rises (ii); that vapour then condenses, releasing heat and dropping the pressure further (iii); the falling pressure finally makes the wind spiral inward, gaining speed (i). Order: (ii) → (iii) → (i).

46. (A) The very low pressure at the centre is exactly what pulls the surrounding higher-pressure air rushing inward, and that inrush is the cyclone's fierce wind. Low pressure drives the wind, it does not stop it.

47. (C) From the centre outward a cyclone has a calm, low-pressure eye, then the violent eye wall, then bands of outer spiralling clouds. The eye and edges are not the violent parts.

48. (B) Air warmed from below expands, becomes lighter and rises away, leaving lower pressure behind. Warm air is less dense, so it does not raise the pressure.

49. (A) The anemometer cups are turned by the wind, so very slow cups mean the wind is light that day. Spin speed does not report temperature, rain or pressure directly.

50. (C) The sea breeze needs the land to be warmer than the sea, which only happens by day when the land heats faster under the Sun. At night that condition is gone (it even reverses), so the daytime sea breeze does not blow at night.

51. (A) Before a cyclone the sea has violent winds and high waves that can capsize fishing boats, so fishermen are warned to stay ashore. The fish, water temperature and "disappearing sea" are not the reason.

52. (D) The fast air inside the dryer's stream has lower pressure than the slower air around it. If the ball drifts to the edge, the higher surrounding pressure pushes it back into the stream, so it hovers and resists being knocked out.

53. (D) Far fewer people die in cyclones today mainly because accurate satellite warnings give time to evacuate to shelters. Cyclones are not weaker, the sea still floods, and the coast is still inhabited.

54. (B) A thunderstorm's rain forms in order: warm moist air rises and cools → its water vapour builds tall clouds → the drops grow heavy and fall as rain. Rain or lightning cannot come first, and cold sinking air would not start it.

55. (B) The Sun's uneven heating of land, sea and atmosphere creates the pressure differences that drive all winds, including the monsoon. The Moon and household fans are not responsible, and ocean currents themselves are ultimately Sun-driven.

56. (A) Cooling the bottle in ice water shrinks the trapped air and drops its pressure below the outside air, so the balloon is sucked inward / collapses into the bottle — the reverse of the hot-water case.

57. (B) As a cyclone intensifies its central pressure falls even lower, and because air rushes harder toward that deeper low, the winds get faster. (Falling pressure with faster — not slower — wind is the key pairing.)

58. (C) A sea breeze needs the land warmer than the sea. If on some afternoon the land is actually cooler than the sea, the required pressure difference is gone or flipped, so the usual sea breeze weakens or even reverses.

59. (D) Fast wind over a roof lowers the pressure above it, creating an upward lift that can tear a loose roof off. Strong, well-anchored roofs resist that lift — which is why coastal homes use them.

60. (B) As the cyclone passes over, a ship first meets the fierce eye-wall winds, then the sudden calm of the eye, then the fierce eye-wall winds again from the opposite direction as the far wall arrives.

Solutions complete: 60 / 60.